2024 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)
July 15-19, 2024. Boston, MA, USA

2024 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) | 979-8-3503-5536-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/AIM55361.2024.10637095

Robot-Assisted Ultrasound Probe Force Control Under Respiration-Induced Motion

Kai-Shiang Yuan[1] and Yu-Hsiu Lee[1]

***Abstract*— A robot-assisted force control system for stable ultrasound imaging has been developed for abdomen intervention. The system aims to integrate 6-DoF robot arm, 6-axis force/torque sensor, and ultrasound probe, featuring automatic stabilization during intervention and real-time compensation for respiratory disturbance. Following the procedural workflow, the system has two operational modes. The first, rooted in admittance control, swiftly positions the robotic-held ultrasound for efficient registration. The second mode, employing the proposed adaptive control, ensures stable contact despite respiratory motion influences, enhancing procedural resilience and effectiveness. The adaptive controller predicts and eliminates disturbances more effectively than the baseline admittance controller, thanks to its utilization of the pseudo-periodic nature of respiration. The efficacy of the adaptive controller has been verified through experiments with a sponge abdomen phantom, demonstrating significant improvements in force regulation.**

I. Introduction

Hepatocellular carcinoma (HCC) ranks as the sixth most prevalent cancer globally, and its incidence is on the rise worldwide, primarily attributed to the escalating prevalence of hepatitis B and C virus infections [1]. According to statistical data from the Taiwan Ministry of Health and Welfare, liver cancer stands as the second leading cause of cancer-related deaths among the Taiwanese population, constituting approximately 15.5% of all cancer-related fatalities [2]. Radiofrequency Ablation (RFA) serves as a primary treatment for early-stage HCC, representing one of the optimal therapeutic choices for patients in whom surgical resection and liver transplantation are deemed unsuitable [3]. During RFA procedures, ultrasound (US) imaging is commonly employed for real-time tracking of tumor positions owing to its cost-effectiveness, portability, and widespread availability [4], [5]. The surgeon faces the challenge of simultaneously holding the US probe in one hand and manipulating the needle with the other during the intervention, highlighting a significant reliance on experienced surgeons as in Fig. 1.

US is a contact-based imaging technique that operates by emitting high-frequency sound waves into the body and detecting their echoes, creating real-time images of internal structures. The stability of the US image during RFA is crucial for several reasons. First and foremost, a stable US image ensures accurate and precise visualization of the targeted tissue or lesion. Moreover, a stable US image is essential for real-time monitoring of the ablation procedure, preventing unintended injury to adjacent structures and organs. It enables medical practitioners to assess the extent of tissue heating and the progress of the ablation in real-time.

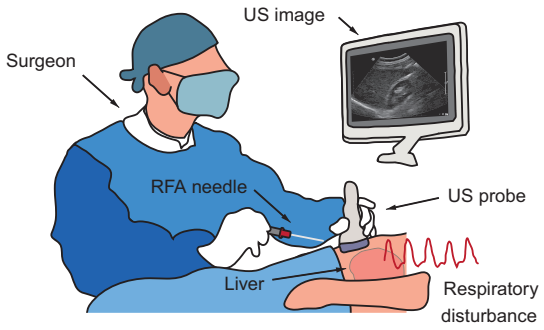


Fig. 1: The schematic of RFA. The surgeon faces the challenge of simultaneously manipulating the needle and US probe during insertion. This task becomes particularly demanding, especially when targeting small lesions or tumors.

Throughout the acquisition process, maintaining proper contact between the US probe and the patient is crucial for enhancing image quality and ensuring patient safety. Excessive contact force carries the risk of deforming the target anatomy and causing harm to the patient, while insufficient force may compromise acoustic coupling, leading to poor image quality. Furthermore, maintaining a stable contact force during the US examination not only enhances the usability and diagnostic capability of the images but also contributes to reducing work-related musculoskeletal disorders among sonographers [6].

With advancements in the accuracy, dexterity, maneuverability, and perceptual capabilities of robots, they can now exert more precise control over the force and position of the US probe. This improvement holds the promise of generating more stable imaging results, consequently alleviating the burden on physicians. As a result, numerous robot-assisted US imaging systems have been developed. Research in this area spans from scanning path tracking to contact force control and image quality optimization, as discussed in a comprehensive overview by Li et al. [7]. The key enabling technology is the contact force control between the US probe and the abdomen.

To achieve contact force control, Gilbertson et al. [6] developed a handheld US probe aimed at eliminating disturbances caused by the sonographer's hand tremors. The

[1] Kai-Shiang Yuan and Yu-Hsiu Lee (corresponding author) are with the Department of Mechanical Engineering, National Taiwan University, Taipei, Taiwan. yuhsiulee@ntu.edu.tw.

This work is supported by the National Science and Technology Council in Taiwan under Grant NSTC 112-2628-E-002-024 and the National Taiwan University Excellence Research Program 113L893901.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 07:36:16 UTC from IEEE Xplore. Restrictions apply.

system integrates a servo motor, accelerometer, load cell, and US probe for active 1-degree-of-freedom (1-DoF) axial motion compensation. However, its handheld robot-assisted operation, demanding physician control and hand-eye coordination, may lead to sensory overloading, posing challenges for effective use in medical settings. For this reason, Ding et al. [8] developed a force-controlled 6-DoF parallel robot to support US-guided interventions. While avoiding the necessity for surgeons to operate the system to obtain US image, the significant volume restricts its range of movement to tissues and organs on the lateral side of the abdomen which may increase the inconvenience during surgery. Fang et al. [9] introduced a force-assisted co-robotic US imaging system utilizing an anthropomorphic 6-DoF robotic arm equipped with two force sensors (a 6-axis force sensor and a 1-axis load cell) integrated with a conventional US probe. The system aims to reduce the force required by the sonographer and provide stable contact.

Existing studies have demonstrated a reduced burden, improved image quality, and enhanced safety through the implementation of robotic technologies. However, it is noteworthy that the dynamic nature of US probe force control in the abdomen remains unaddressed. Due to its position beneath the diaphragm, the liver is more susceptible to the influence of respiratory motion compared to other organs. Standard admittance control methods may prove insufficient in such cases. As such, herein we propose the use of a force-controlled robotic platform which not only improves the procedural flow but also ensures a stable contact force between the US probe and the patient's abdomen under respiratory disturbance.

Fig. 2 depicts the force trajectory of the proposed robot-assisted procedure for US probe force control: During the calibration phase, the F/T sensor compensates for the gravity exerted by the end-effector. Transitioning to the free motion and registration phase, the initial registration mode is activated, signifying contact with the abdomen. In the adaptation phase, the adaptive controller is implemented to counteract respiratory disturbances. By leveraging the pseudo-periodic nature of breathing characteristics [10], specialized adaptive feedback control adjusts in real-time to changes in the frequency components of respiratory disturbances caused by the patients, optimizing controller parameter adjustments. In contrast to model reference adaptive control or self-tuning adaptive control, where control parameters adapt to variations in plant dynamics, the employed adaptive controller herein assumes knowledge of plant dynamics to minimize the impact of unknown external input signals on regulated output signals. Stability is warranted so long as the plant model is sufficiently close to the actual system [11]. Preliminary experimental results on a phantom target with comparable mechanical properties demonstrate the merits of the approach.

The main contributions of this article are as follows:

- Automatic contact force control between the US probe and patient's abdomen without surgeon manipulation during the US acquisition.
- Real-time alleviation of disturbance caused by respiratory motion enhances US image quality and improves patients' comfort.

The rest of this article is organized as follows. Section II briefs the procedure of the US intervention and outlines the proposed two modes: Initial registration mode and respiratory disturbance rejection mode. Section III delves into the hardware configuration of the system, gravity compensation algorithm, and the design of admittance and adaptive controllers. Section IV evaluates the performance of the proposed controller in comparison to the baseline admittance control system. Finally, Section V concludes the article and highlights future directions.

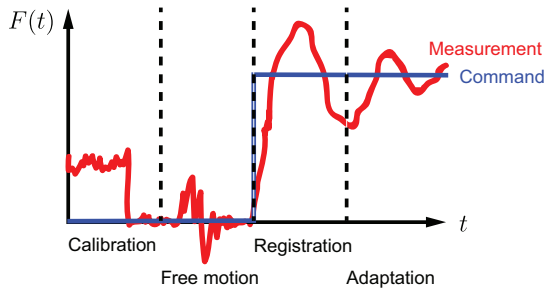


Fig. 2: Illustrative time evolution of the proposed robot-assisted procedure for US probe force control.

II. Robot-Assisted Ultrasound Intervention Procedure

The procedure of US-guided intervention can be categorized into following key steps: pre-operative imaging, registration, insertion, ablation, and post-operative evaluation. This is illustrated in Fig. 3, associated with the relevant robot-assisted procedures in this work. Before the intervention, surgeons may perform pre-procedure imaging, such as Magnetic Resonance Imaging (MRI) or Computed Tomography (CT), to assess the severity of the illness or plan the insertion path. Subsequently, the registration phase establishes the connection between the US probe and the patient. Surgeons carefully manipulate the probe into the Region of Interest (ROI) to ensure that the US image quality is sufficient for executing the pre-procedure insertion plan. Throughout the insertion and ablation phases, the US probe continuously monitors the procedure to ensure alignment with the plan, confirming precise targeting of the tumor or lesion. Finally, post-procedure evaluation relies on high-resolution MRI or CT images to verify the complete removal of the tumor or lesion.

Therefore, the robot-assisted force-controlled US imaging system can be structured into two modes: the initial registration mode and the respiratory disturbance rejection mode. In the initial registration mode, the surgeon manually manipulates the system by holding the probe and positioning the US probe based on their experience to obtain an optimal

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 07:36:16 UTC from IEEE Xplore. Restrictions apply.

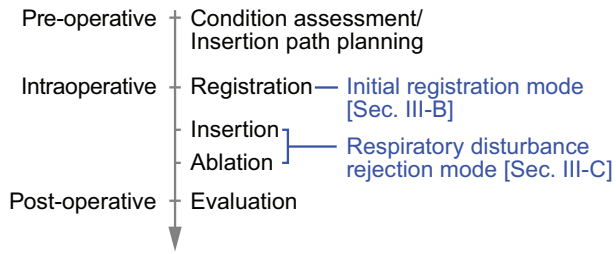


Fig. 3: RFA process and the corresponding robot-assisted mode. The blue text highlights the designed mode intended to assist with the procedure.

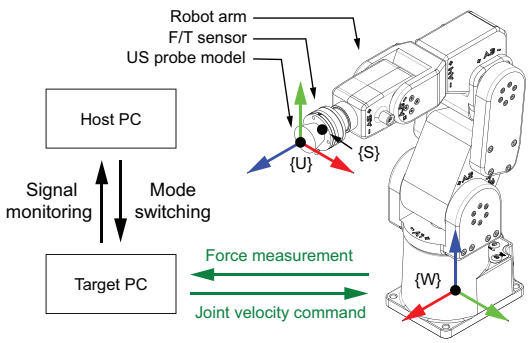


Fig. 4: The robot-assisted force-controlled US imaging system and mechatronic system architecture. The system comprises three components: 6-axis Robot arm, 6-DoF F/T sensor, and US probe model. {U}, {S} and {W} denote the reference frames for the US probe model, the F/T sensor, and the world, respectively. The green arrow line below indicates the EtherCAT protocol.

US image. In this mode, all 6-DoF are active, allowing the surgeon to freely determine the probe's position and orientation. Once the US probe is appropriately adjusted and the reliability of the US image is confirmed, the respiratory disturbance rejection mode can be activated. In this mode, the surgeon no longer needs to control the system, and the system is limited to 1-DoF in the direction normal to the patient's abdomen. At this stage, the respiratory disturbance compensation signal calculated by the adaptive feedback controller is added to the system's input, facilitating constant force control. This allows the surgeon to focus more on the insertion process, improving precision and overall procedure efficiency.

III. Hardware Configuration and Control Structure

The robot-assisted force-controlled US imaging system comprises three main components (Fig. 4). Firstly, a 6-DoF robot manipulator (Meca500 R3, Mecademic Inc., Canada) mimics the surgeon's movements. Secondly, a 6-axis Force/Torque (F/T) sensor (ATI-Mini40-E) is attached to the end of the manipulator to measure contact force. This measurement is utilized in both admittance control and adaptive control. Lastly, the US probe model is connected after the F/T sensor.

In the following section, we will briefly discuss how we compensate for the initial bias and gravity measurement caused by the end-effector and the F/T sensor itself. We will then introduce the admittance controller and adaptive feedback controller sequentially for the registration mode and the disturbance rejection mode, respectively.

A. Gravity compensation of end-effector

Prior to system operation, we compensate for the gravity of the US probe through a set of properly chosen orientations [12]. The basic concept of this method is to construct augmented matrices of unknown bias and gravity effects, then solve them using standard least-squares. These unknowns are related by Eq. 1:

$$\begin{cases} {}^S\boldsymbol{f} = {}^S\boldsymbol{f}_0 + {}^S\boldsymbol{f}_g \\ {}^S\boldsymbol{\tau} = {}^S\boldsymbol{\tau}_0 + {}^S\boldsymbol{\tau}_g \\ {}^S\boldsymbol{\tau}_g = {}^S\boldsymbol{r} \times {}^S\boldsymbol{f}_g \\ {}^S\boldsymbol{f}_g = {}^S_W\boldsymbol{R} \cdot m\,{}^W\boldsymbol{g} \end{cases} \tag{1}$$

where ${}^S\boldsymbol{f}$ and ${}^S\boldsymbol{\tau}$ are the F/T sensor measurements, ${}^S\boldsymbol{f}_0$ and ${}^S\boldsymbol{\tau}_0$ are the initial biases, ${}^S\boldsymbol{f}_g$ and ${}^S\boldsymbol{\tau}_g$ are the force and torque due to gravity of the end-effector and F/T sensor itself, ${}^S\boldsymbol{r}$ is the mass center of the combination of end-effector and F/T sensor, ${}^S_W\boldsymbol{R}$ represents the rotational matrix from world reference frame {W} to F/T sensor reference frame {S}, m denotes the mass of the combination of F/T sensor and the end-effector, and ${}^W\boldsymbol{g}$ indicates the gravitational acceleration in {W}. An associated advantage is that this approach eliminates the need for prior knowledge and pre-input of the physical parameters of the US probe, enhancing the convenience of integrating with existing US devices during installation.

B. Initial registration mode

To enhance the intuitiveness and convenience of the registration step, we employ an admittance controller. Admittance control is a widely utilized method in physical human-robot interaction (pHRI) [13]. By defining the virtual dynamics, which represent the relationship between the measured force and the reference velocity command, we can manipulate the apparent dynamics of the entire system according to our desired specifications [14]. For simplicity, we configured the admittance controller C_{adm} to provide pure damping, specifically accommodation control. In this setup, the controller acts as a constant gain. The value of C_{adm} has been set to 160 N·sec/m through tuning and testing. In practical applications, the operator has the flexibility to adjust this value according to personal preferences. Moreover, the coupled stability is relatively high compared to other combinations of haptic elements, indicating increased safety during pHRI.

For robot manipulation, we choose for joint velocity commands instead of position or end-effector attitude commands. This choice provides a more straightforward representation of the power conjugation between force and velocity. It also avoids the potential singularity caused by end-effector atti-

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 07:36:16 UTC from IEEE Xplore. Restrictions apply.

tude commands. The task-space velocity vector with respect to the US probe reference frame ${}^{U}\dot{\boldsymbol{p}}_U$ can be denoted as follows:

$$ {}^{U}\dot{\boldsymbol{p}}_U = \begin{bmatrix} \dot{x} & \dot{y} & \dot{z} & \dot{\theta}_x & \dot{\theta}_y & \dot{\theta}_z \end{bmatrix}^T \tag{2} $$

The relationship between ${}^{U}\dot{\boldsymbol{p}}_U$ and the joint-space velocity vector $\dot{\boldsymbol{\theta}}$ can be derived:

$$ \begin{aligned} \dot{\boldsymbol{\theta}} &= \begin{bmatrix} \dot{\theta}_1 & \dot{\theta}_2 & \dot{\theta}_3 & \dot{\theta}_4 & \dot{\theta}_5 & \dot{\theta}_6 \end{bmatrix}^T \\ &= \boldsymbol{J}^{-1} {}^{W}\dot{\boldsymbol{p}}_U \\ &= \boldsymbol{J}^{-1} {}^{W}_{U}\boldsymbol{T}\, {}^{U}\dot{\boldsymbol{p}}_U \end{aligned} \tag{3} $$

where

$$ \boldsymbol{J}^{-1} = \begin{bmatrix} \dfrac{\partial {}^{W}\boldsymbol{p}_U}{\partial \theta_1} & \dfrac{\partial {}^{W}\boldsymbol{p}_U}{\partial \theta_2} & \cdots & \dfrac{\partial {}^{W}\boldsymbol{p}_U}{\partial \theta_6} \end{bmatrix} \tag{4} $$

represents the Jacobian matrix and ${}^{W}_{U}\boldsymbol{T}$ represents the transformation matrix from frame {U} to frame {W}. The architecture of the admittance control for initial registration mode is shown in Fig. 5a and is denoted by $\boldsymbol{P}$, serving as the baseline performance.

C. Respiratory disturbance rejection mode

While the admittance controller is simple, its performance is limited. To enhance the baseline admittance controller, adaptive inverse control techniques are employed [15]. This design, leveraging prior knowledge of the system model, offers excellent disturbance-canceling capabilities without concerns about stability issues [11].

The employed adaptive feedback control structure is shown in Fig. 5b. The adaptation is recursively minimizing the following cost function:

$$ \min_{C_{fb}} \left\| (1 - C_{fb}\hat{P})e \right\|_2 \tag{5} $$

where $\hat{P}$ represents the model of the admittance control system $\boldsymbol{P}$ in the z direction, as illustrated in Fig. 5a; C_{fb} is the adapted finite impulse response (FIR) filter, which minimizes the estimated disturbance e. The adaptive algorithm in effect constructs the inverse dynamics C_{fb} best suited for the frequency components of the unknown disturbance signal e [10]. If the disturbance contains certain periodic components, the adaptive controller can predict and generate the required control signal for disturbance cancellation.

For the adaptation algorithm, recursive least squares (RLS) is chosen for faster convergence [16]. The RLS algorithm is presented in Algorithm 1, where $s[n]$ is the desire response, $\boldsymbol{u}[n]$ is the tap-input vector, $\hat{\boldsymbol{w}}[n]$ is the estimate tap-weight vector of C_{fb}, $\boldsymbol{\Phi}^{-1}[n]$ is the inverse correlation matrix of $\boldsymbol{u}[n]$, δ is the regulation parameter which is assigned based on the signal-to-noise ratio (SNR). It should be assigned a small value for high SNR and a large value for low SNR, λ is the forgetting factor which is an exponentially weighted factor (≤ 1) that discounts the importance of past input/output (I/O) data. It provides adaptation capability when the fundamental harmonic of the respiratory disturbance experiences a slow frequency shift, $\boldsymbol{k}[n]$ is the gain vector which updated by $\boldsymbol{k}[n] = \boldsymbol{\Phi}^{-1}[n]\boldsymbol{u}[n]$, $\xi[n]$ is the *a priori* estimation error, the asterisk denotes complex conjugation, and the superscript T denotes transposition. Corresponding to Fig. 5b, the desired response is $s[n] = e[n]$, and the input signal is $u[n] = \hat{P}e[n]$.

Algorithm 1: RLS algorithm

Input: Desire response $s[n]$, tap-input vector $\boldsymbol{u}[n]$.
Output: Estimate tap-weight vector $\hat{\boldsymbol{w}}[n]$.

Initialize the algorithm by setting
$\hat{\boldsymbol{w}}[0] = 0$,
$\boldsymbol{\Phi}^{-1}[0] = \delta^{-1} I$,
where

$$ \delta = \begin{cases} \text{small positive constant, for high SNR} \\ \text{large positive constant, for small SNR} \end{cases} . $$

For each instant of $n = 1, 2, 3, ...$, compute

$$ \boldsymbol{k}[n] = \frac{\lambda^{-1}\boldsymbol{\Phi}^{-1}[n-1]\boldsymbol{u}[n]}{1 + \lambda^{-1}\boldsymbol{u}^T[n]\boldsymbol{\Phi}^{-1}[n-1]\boldsymbol{u}[n]}, $$

$\xi[n] = s[n] - \hat{\boldsymbol{w}}^T[n-1]\boldsymbol{u}[n]$,
$\hat{\boldsymbol{w}}[n] = \hat{\boldsymbol{w}}[n-1] + \boldsymbol{k}[n]\xi^*[n]$,
where $\boldsymbol{\Phi}^{-1}[n] = \lambda^{-1}\boldsymbol{\Phi}^{-1}[n-1] - \lambda^{-1}\boldsymbol{k}[n]\boldsymbol{u}^T[n]\boldsymbol{\Phi}^{-1}[n-1]$.

IV. Experimental Results

In this section, we will initially model the integrated dynamics of the admittance controlled robot and the target phantom, which represents the patient's abdomen. Once the model is established, the adaptive feedback controller can be constructed. Finally, an experiment is conducted to verify the effectiveness of the adaptive feedback controller in rejecting respiratory disturbance.

The experimental setup is depicted in Fig. 6. The robot-assisted US imaging system is controlled by an NI LabVIEW Real-Time Target (Intel Core i7-4790 Processor) through the EtherCAT protocol. Both the admittance control and adaptive controller are executed at a sampling rate of 100 Hz. The abdomen phantom, made of sponge to mimic the physical properties of the patient's abdomen, is used. The respiratory disturbance is artificially added in software at the junction of force measurement. In the future, the disturbance will be generated using the linear actuator presented.

A. Unit Impulse Response of the Robot and Patient System

For the deployment of the model-based adaptive controller, we first require a model of the overall robot and patient dynamics. We achieve this by subtracting two force step response input-output (I/O) data with a one-step delay and normalizing the amplitude to obtain the unit impulse response of the system. Before executing the step command, the US probe model is manually moved to establish contact with the phantom. Subsequently, the US probe model is set to maintain a contact force of 1 N with the phantom, as the system includes the interaction between the US probe model and the abdomen phantom. Then, for better signal-to-noise ratio, the robot is commanded to regulate at 4 N, as depicted

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 07:36:16 UTC from IEEE Xplore. Restrictions apply.

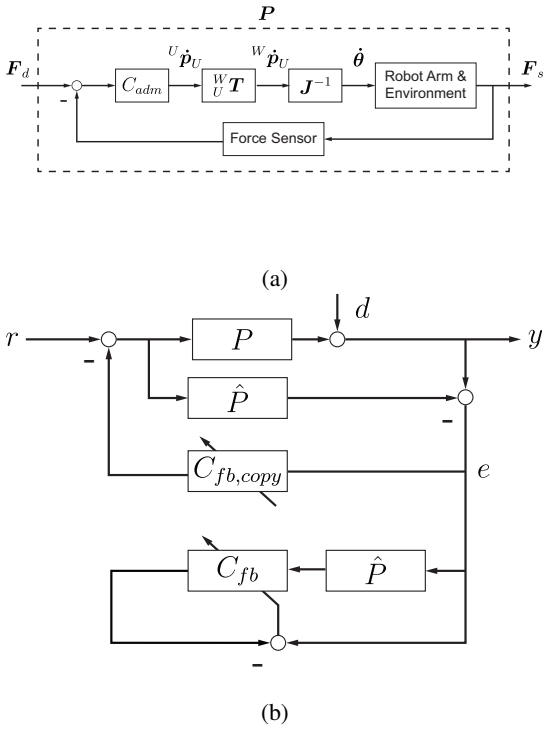


Fig. 5: (a) Baseline admittance control: All six elements in desire force vector F_d are set to zero during the registration mode in order to interact with the surgeon. F_s denotes the measurement value of the F/T sensor. (b) Add-on adaptive control: the system P is the admittance-controlled robot in z direction, $\hat{P}$ the identified model, r the constant reference force, d the respiratory output disturbance, y the contact force, and e the estimated disturbance between the plant output and model output.

in the upper half of Fig. 7. Taking into account the variability in abdominal composition among individuals, which can affect system dynamics, we conduct multiple experiments to gather ensemble average data for constructing the nominal FIR model, as shown in the bottom half of Fig. 7.

B. Respiratory Disturbance Rejection

The experiment procedure is similar to the previous one. Before sending the step command into the system, the US probe model makes contact with the abdomen phantom manually. After the step command is executed, a respiratory-like disturbance and the adaptive feedback signal are added to the system.

The respiratory-like disturbance is constructed by combining a sinusoidal disturbance and a band-limited stochastic disturbance (Fig. 8). The sinusoidal disturbance is configured to mimic the main frequency component of the respiratory disturbance, with a frequency set to 0.3 Hz (about 18 breaths/min) [17]. The band-limited stochastic disturbance is added to make the disturbance more closely resemble real-life conditions. The bandwidth of the band-pass filter is set to 0.6 $\sim$ 0.8 Hz.

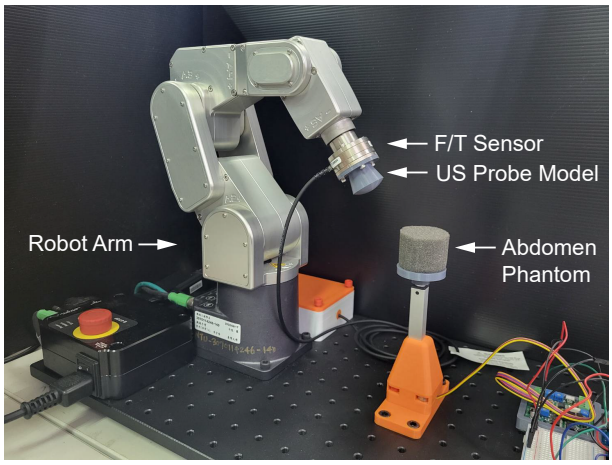


Fig. 6: The experimental setup. The contact between the US probe model and the abdomen phantom is established manually before the experiment.

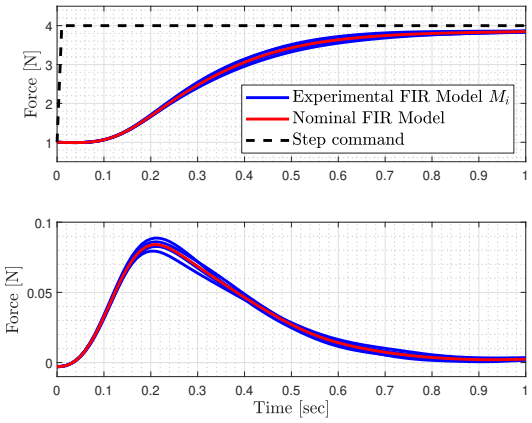


Fig. 7: The step response (top) and the unit impulse response (bottom) of the integrated dynamics of the robot and patient system. The red line represents the nominal model constructed using the ensemble average.

The adaptive feedback signal is multiplied by a ramping function that gradually opens after 20 seconds to avoid large transients and saturation during adaptation. To evaluate the effectiveness of the adaptive feedback controller, the results are compared with those without the adaptive feedback controller, namely only the baseline admittance controller is in effect. This is shown in Fig. 9, and the RMSE of the steady-state ($t \geq 25$ s) output with and without adaptive feedback controller is 0.381 and 0.7318 N, respectively. These results show the adaptive feedback controller improves the system in terms of the steady-state RMSE.

V. Conclusion and Future Works

A force-controlled robot-assisted US imaging system has been developed to support US-guided abdominal intervention. The system integrates a 6-axis robot arm, a 6-DoF F/T sensor, and a US probe, offering advantages such as a small volume, compatibility with existing US devices,

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 07:36:16 UTC from IEEE Xplore. Restrictions apply.

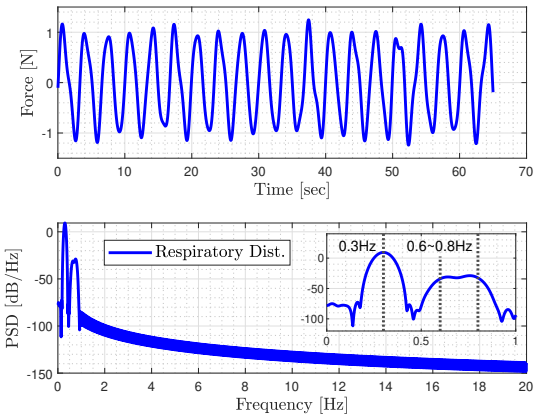


Fig. 8: The pre-defined time domain signal and power spectral density (PSD) of the respiratory-like disturbance.

	w/ Adaptive FB	w/o Adaptive FB
Max Error [N]	0.5265	1.2134
Mean Error [N]	-0.0066	0.0036
RMS Error [N]	0.1458	0.7113

TABLE I: Contact force error statistics of steady-state (calculated using $t \geq 25$ s data in Fig. 9)

automatic stabilization during intervention, and real-time respiratory disturbance rejection. The system operates in two modes in accordance with the current clinical practice: initial registration mode and respiratory disturbance rejection mode. The former provides an intuitive way for surgeons to register the system, and the latter allows for individual execution of constant force control, enabling surgeons to focus more on the insertion. Experimental results with a sponge abdomen phantom demonstrate significant improvement compared to the open-loop system. In the future, practical enhancements will be made by integrating the system with existing US probes and experimenting with an actuating gelatin abdomen phantom. The deterministic and stochastic components of the disturbance may be handled separately to enhance overall performance. Evaluations will be based on the quality of the US image itself, primarily focusing on safety considerations for both the patient and surgeon.

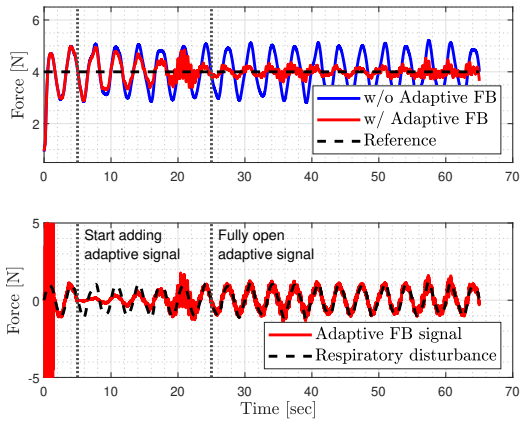


Fig. 9: The experimental result (top) and the feedback signal (bottom) of adaptive feedback control for respiratory disturbance rejection. The set-point is set to 4 N based on the typical value of conducting US imaging.

REFERENCES

[1] R. Lencioni and L. Crocetti, "Radiofrequency ablation of liver cancer," *Techniques in vascular and interventional radiology*, vol. 10, no. 1, pp. 38–46, 2007.

[2] M. of Health and Welfare. (2021) Eighty percent of liver cancer is associated with hepatitis b and c. screening and treatment are crucial! protect your liver! [Online]. Available: https://www.mohw.gov.tw/cp-5020-63007-1.html

[3] S. McDermott and D. A. Gervais, "Radiofrequency ablation of liver tumors," in *Seminars in interventional radiology*, vol. 30, no. 01. Thieme Medical Publishers, 2013, pp. 049–055.

[4] J. W. Kim, S. S. Shin, S. H. Heo, J. H. Hong, H. S. Lim, H. J. Seon, Y. H. Hur, C. H. Park, Y. Y. Jeong, and H. K. Kang, "Ultrasound-guided percutaneous radiofrequency ablation of liver tumors: how we do it safely and completely," *Korean journal of radiology*, vol. 16, no. 6, pp. 1226–1239, 2015.

[5] Y. J. Kim, M. W. Lee, and H. S. Park, "Small hepatocellular carcinomas: ultrasonography guided percutaneous radiofrequency ablation," *Abdominal imaging*, vol. 38, pp. 98–111, 2013.

[6] M. W. Gilbertson and B. W. Anthony, "Force and position control system for freehand ultrasound," *IEEE Transactions on Robotics*, vol. 31, no. 4, pp. 835–849, 2015.

[7] K. Li, Y. Xu, and M. Q.-H. Meng, "An overview of systems and techniques for autonomous robotic ultrasound acquisitions," *IEEE Transactions on Medical Robotics and Bionics*, vol. 3, no. 2, pp. 510–524, 2021.

[8] J. Ding, D. Swerdlow, S. Wang, E. Wilson, J. Tang, and K. Cleary, "Robotically assisted ultrasound interventions," in *Medical Imaging 2008: Visualization, Image-Guided Procedures, and Modeling*, vol. 6918. SPIE, 2008, pp. 746–753.

[9] T.-Y. Fang, H. K. Zhang, R. Finocchi, R. H. Taylor, and E. M. Boctor, "Force-assisted ultrasound imaging system through dual force sensing and admittance robot control," *International journal of computer assisted radiology and surgery*, vol. 12, pp. 983–991, 2017.

[10] Y.-H. Lee, X. Li, J. Simonelli, D. Lu, H. H. Wu, and T.-C. Tsao, "Adaptive tracking control of one-dimensional respiration induced moving targets by real-time magnetic resonance imaging feedback," *IEEE/ASME Transactions on Mechatronics*, vol. 25, no. 4, pp. 1894–1903, 2020.

[11] P. K. Orzechowski, N. Y. Chen, J. S. Gibson, and T.-C. Tsao, "Optimal suppression of laser beam jitter by high-order rls adaptive control," *IEEE Transactions on Control Systems Technology*, vol. 16, no. 2, pp. 255–267, 2008.

[12] S. Vougioukas, "Bias estimation and gravity compensation for force-torque sensors," in *Proceedings of International Conference on Mathematical Methods and Computational Techniques in Electrical Engineering*. Citeseer, 2001, pp. 82–85.

[13] F. Dimeas and N. Aspragathos, "Online stability in human-robot cooperation with admittance control," *IEEE transactions on haptics*, vol. 9, no. 2, pp. 267–278, 2016.

[14] A. Q. Keemink, H. van der Kooij, and A. H. Stienen, "Admittance control for physical human–robot interaction," *The International Journal of Robotics Research*, vol. 37, no. 11, pp. 1421–1444, 2018.

[15] B. Widrow and E. Walach, *Adaptive inverse control: A signal processing approach*. John Wiley & Sons, 2008.

[16] S. Haykin, "Adaptive filter theory," *prentice hall google schola*, vol. 2, pp. 286–292, 2002.

[17] E. D. Brandner, A. Wu, H. Chen, D. Heron, S. Kalnicki, K. Komanduri, K. Gerszten, S. Burton, I. Ahmed, and Z. Shou, "Abdominal organ motion measured using 4d ct," *International Journal of Radiation Oncology* Biology* Physics*, vol. 65, no. 2, pp. 554–560, 2006.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 07:36:16 UTC from IEEE Xplore. Restrictions apply.